

Quiz 7

● Graded

Student



Total Points

15 / 30 pts

Question 1

(no title)

5 / 10 pts

+ 10 pts Correct

+ 7 pts on the right track

✓ + 5 pts Click here to replace this description.

+ 3 pts there is a good idea here but it is incorrect or barely fleshed out

+ 2 pts honest attempt

+ 0 pts left blank

Question 2

(no title)

7 / 10 pts

+ 10 pts Correct

✓ + 7 pts Correct but explanation is unclear or shows poor understanding

+ 0 pts Incorrect answer and justification

Question 3

(no title)

3 / 10 pts

+ 10 pts Correct

+ 7 pts on the right track

+ 5 pts answer shows some understanding

✓ + 3 pts Honest attempt

+ 0 pts Blank

Math 54 Discussion Worksheet

Name: 

ID: _____

Problem 1

Prove that if A is a real $n \times n$ matrix, then its complex eigenvalues come in conjugate pairs, i.e. show that if λ is an eigenvalue, then $\bar{\lambda}$ is also an eigenvalue.

If λ has the representation of $z = \operatorname{Re} z + i \operatorname{Im} z$, i.e.

$z = a + bi \rightarrow$ it is the case that for any matrix $A \in \mathbb{R}^{n \times n}$, can be expressed as $\overline{a + bi} \vee a + bi$ when writing it as a linear combination. i.e., you don't have to show it both ways.

$$\lambda = \bar{\lambda} \Rightarrow A\lambda = A^{-1}\lambda \Rightarrow \lambda =$$

Problem 2 $c_i = \frac{y \cdot u_i}{u_i \cdot u_i}$, $y - \tilde{y} \rightarrow \sqrt{a^2 + b^2}$, $v_1 \cdot v_1 = 1 \wedge v_1 \cdot u_1 = 0$

If the vectors in an orthogonal set of non-zero vectors are normalized, then some of the resulting vectors may no longer be orthogonal to each other.

False, orthogonality has the conditions $\forall v_i, v_j \in W$ as a basis, then $W = c_1 v_1 + \dots + c_n v_n = 0$, meaning they are the same norm (length), thus that doesn't change

Problem 3

Recall that we define the distance between two vectors $u, v \in \mathbb{R}^n$ as $d(u, v) = \|u - v\|$. Prove the Pythagorean Theorem: two vectors are orthogonal if and only if $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

$$\left(\sqrt{a_1^2 + \dots + a_n^2} \quad \forall \text{ matrix } A \in \mathbb{R}^{n \times n} \right)$$

To find orthogonality for any matrix $u \sim v$, we have the invariant that their ^{correction is for orthonormal} unit vectors must be 1, the inner product of the $\langle u + v \rangle$ must be 0, because they are perpendicular.

$$\|u - v\| \rightarrow (\forall u_i, v_i \in u, v): \sqrt{u_1^2 + \dots + u_n^2} + \sqrt{v_1^2 + \dots + v_n^2}$$

$$\rightarrow \sqrt{u_1^2 + \dots + u_n^2} + \sqrt{v_1^2 + \dots + v_n^2} = \sqrt{\|u\|^2 + \|v\|^2} = \|u + v\|^2 \wedge$$

$$= \|u\|^2 + \|v\|^2$$